Health Data Science

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Reproducible Reporting in Practice: developing confidence through a practical and encouraging introduction

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Why reproducibility is key in health data research analysis

Reproducibility turns an analysis from a one-off result into evidence others can inspect, rerun, and trust.

Because health data analysis carries clinical, public, and governance consequences.

Traceability
Every number can be linked back to data processing choices, definitions, and code.

Quality & error reduction
Rerunnable workflows make mistakes easier to detect before results influence decisions.

Collaboration across teams
Analysts, clinicians, supervisors, and reviewers can work from the same evidence trail.

Trustworthy translation
Transparent methods strengthen credibility when findings move into policy, practice, or publication.

Aim of this session is to take you from
“I am not a coder” or “I don’t know this programming language”

To feeling confident enough

To work with free text code
Understand what the code is doing and
Evaluate the code in your own context

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1 0–10 min
Why it matters

2 10–25 min
Hands-on Anatomy of a report

3 25–50 min
Hands-on clinical EDA

4 50–60 min
Reflect & next steps

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Approach

Clinical question

↓

Data decision

Real-world survey data

↓

Code & report

Practical
Reproducible

↓

Evidence for care

Refine

FINER

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PICO model

PICO

- **P** — Population or patient
- **I** — Intervention
- **C** — Comparison or Control
- **O** — Outcome

@Fatemehtrb

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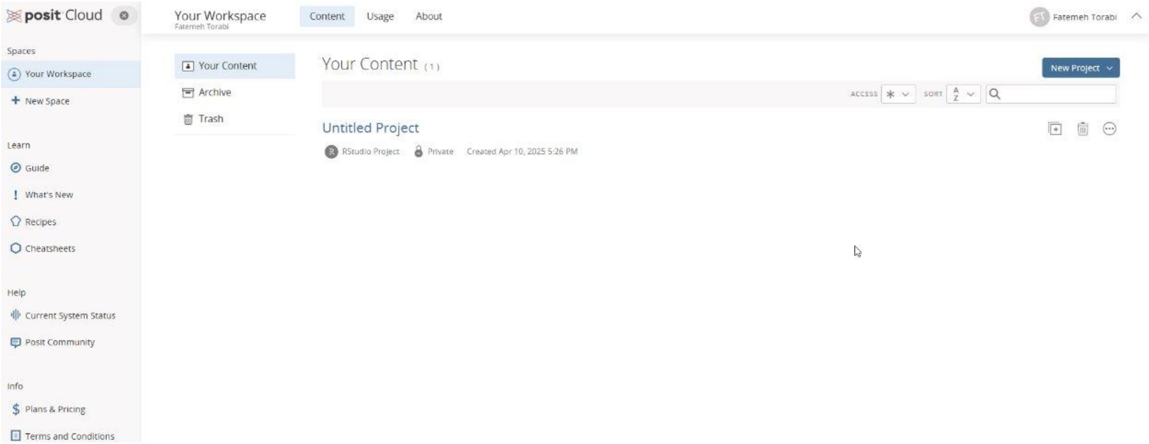
Getting started

https://posit.cloud/



[Create your posit cloud account and explore](https://posit.cloud/)

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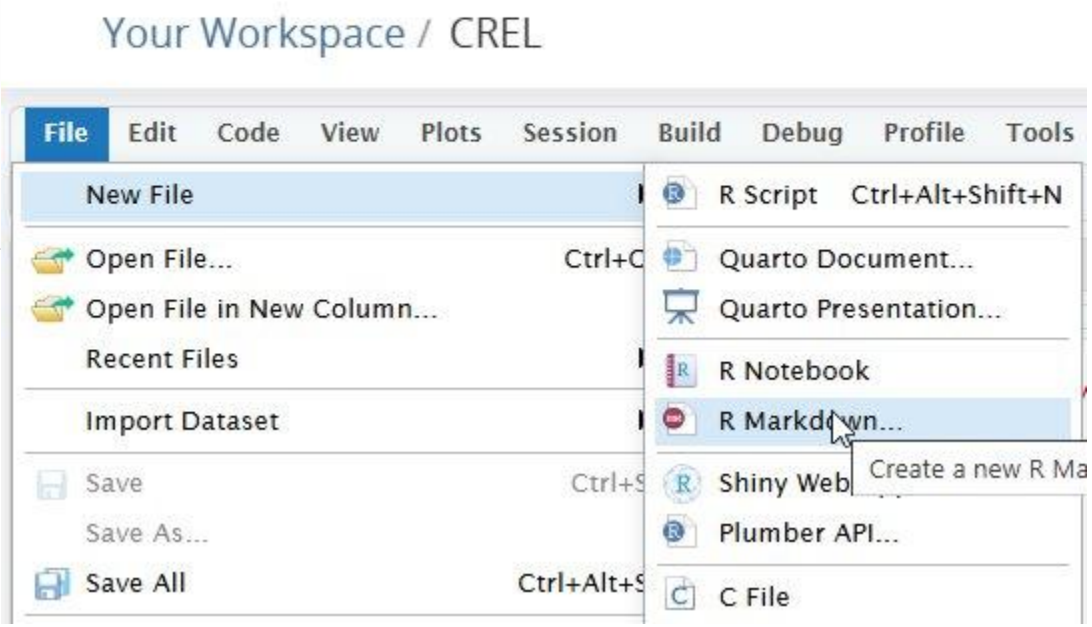
Live coding demo



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Exploratory data analysis on a clinical dataset

Context

Diabetes Risk Factors

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A growing burden, hiding in plain sight

Type 2 diabetes is largely driven by modifiable factors, yet the condition keeps spreading. This brief asks which characteristics are independently associated with diabetes in a nationally representative U.S. sample.

Source: NHANES, open-source national survey data

ADULTS LIVING WITH DIABETES WORLDWIDE

537M IDF, 2021

AMERICANS AFFECTED

37.3M ~11.3% of the population

ANNUAL U.S. COST OF DIAGNOSED DIABETES

$327B 2017 estimate

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Live coding demo

Go to this github repository

Evidence

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Four factors move the odds

A logistic regression isolates each factor while holding the others constant. Three independently raise the odds of diabetes; higher household income lowers them.

Racial and ethnic disparities persist even after adjusting for these factors.

Age	Odds climb steadily with each additional year	Risk ↑
BMI	Higher body mass index is strongly associated	Risk ↑
Inactivity	Physical inactivity raises the odds independently	Risk ↑
Income	Higher household income is protective	Risk ↓

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Thank you!